Research Statement

Asikul Islam

Department of Mathematics, University of Houston

Research Overview

My research lies at the intersection of numerical partial differential equations (PDEs), reduced-order models (ROMs), PDE-constrained optimization, inverse problems, uncertainty quantification (UQ), and scientific computing. I am particularly interested in computational problems in which high-fidelity PDE models accurately represent complex physical or biological processes but become prohibitively expensive when they must be evaluated repeatedly for parameter estimation, optimization, uncertainty analysis, or statistical inference.

My doctoral research, entitled *Tensorial Reduced-Order Models for Parameter Estimation in Brain Tumor Growth Models*, develops computational methods for overcoming this many-query bottleneck. Brain tumor growth provides a particularly challenging setting: the governing models are nonlinear, heterogeneous, time dependent, and parameterized by quantities that cannot usually be measured directly. At the same time, parameter estimation requires repeated numerical solution of the forward problem together with sensitivity or adjoint computations.

My research addresses these challenges through a sequence of closely connected problems:



A central theme throughout my work is that computational acceleration should not come at the expense of mathematical consistency. I therefore seek reduced models and optimization methods that preserve a transparent connection to the underlying PDE and to the numerical discretization used to solve it.

Tensorial Reduced-Order Modeling for Brain Tumor Growth

The first major component of my doctoral research concerns efficient surrogate models for nonlinear parametric reaction–diffusion systems arising in brain tumor growth modeling. I consider models of increasing complexity, beginning with a scalar single-species reaction–diffusion equation and extending to a six-state, nine-parameter multi-species go-or-grow model. The multi-species system represents interacting proliferative, infiltrative, and necrotic tumor populations together with oxygen and tissue-related variables.

The governing PDE systems are discretized using finite-volume methods on heterogeneous brain domains. For the time-dependent problem, operator-splitting strategies separate diffusion from nonlinear reaction mechanisms, allowing the individual components to be treated with appropriate numerical methods while providing a convenient structure for reduced-order modeling.

To accelerate repeated simulations, my collaborators and I developed tensorial ROMs based on the Higher-Order Singular Value Decomposition (HOSVD) in Tucker format and the Tensor Train (TT) decomposition. Unlike a conventional snapshot matrix, the tensor representation retains the natural organization of the numerical solution with respect to space, time, and each model parameter. This allows low-rank structure to be identified simultaneously across several parameter directions.

We developed both intrusive and non-intrusive tensorial formulations. In the intrusive formulation, the compressed tensor representation is used to construct parameter-dependent reduced spaces and the governing discrete equations are projected onto these spaces. In the non-intrusive formulation, approximate solution trajectories are obtained directly from the tensor representation without carrying out a reduced time integration. A classical Proper Orthogonal Decomposition (POD) reduced model was used as a reference approach for evaluating the tensorial methods.

A major computational challenge in high-dimensional parametric models is the size of the snapshot data required during the offline stage. To address this issue, we introduced spatial pre-compression before performing the higher-order tensor decomposition. This substantially reduces memory requirements and makes tensor construction feasible for substantially larger snapshot ensembles.

Our numerical results demonstrate online speedups of approximately 85×–120× relative to the corresponding high-fidelity full-order model (FOM) simulations while maintaining high approximation accuracy. These results establish tensorial ROMs as an effective computational foundation for many-query tasks in nonlinear parameter-dependent PDE systems.

This work is presented in

> *Tensorial Reduced-Order Models for Parametric Coupled Reaction-Diffusion Systems: Application to Brain Tumor Growth Modeling*
>
> A. Islam, Md R. Bin Mizan, M. Olshanskii, and A. Mang
>
> arXiv:2603.14101, under review.

The development of these fast forward surrogates naturally led to the next question in my research: whether the same reduced structure can be used reliably for solving inverse problems.

PDE-Constrained Optimization and Parameter Estimation

My second research direction focuses on estimating unknown physical and biological parameters from observations of tumor evolution. Parameters controlling diffusion, proliferation, phenotype transitions, necrosis, oxygen supply, and oxygen consumption are typically not known directly. Recovering them leads to a nonlinear PDE-constrained optimization problem.

I first studied parameter estimation for a single-species reaction–diffusion tumor model. For this problem, I formulated a regularized least-squares inverse problem and derived the state sensitivity equation, adjoint equation, reduced gradient, first-order optimality system, and second-order information required by Newton-type optimization algorithms.

I subsequently extended the framework to the nonlinear multi-species system, where nine interacting physical parameters are estimated simultaneously. The substantially higher dimensionality of the parameter space, nonlinear coupling of the state variables, parameter constraints, and heterogeneous diffusion make this inverse problem significantly more challenging.

A central methodological principle of this work is *discretize-then-optimize*. Rather than deriving derivatives of the continuous PDE and subsequently discretizing them, I differentiate the fully discrete numerical map used by the forward solver. Consequently, the sensitivity, adjoint, gradient, and second-order information supplied to the optimization algorithm correspond directly to the numerical model used to evaluate the objective functional.

For the multi-species problem, the forward solver uses a second-order Strang-splitting structure in which Crank–Nicolson diffusion half-steps surround a second-order Runge–Kutta reaction step. I

derive the corresponding discrete sensitivities and adjoint relationships and use them to construct derivatives of the reduced objective with respect to all nine model parameters.

My numerical work has investigated several optimization algorithms, including projected gradient methods, accelerated first-order methods, limited-memory quasi-Newton methods, and Gauss–Newton approaches. My current framework emphasizes a projected Gauss–Newton method that exploits the nonlinear least-squares structure of the parameter-estimation problem. Physical box constraints and the ordering constraints associated with oxygen thresholds are enforced throughout the optimization.

Regularization is particularly important because tumor-growth inverse problems may be ill-conditioned and individual parameters can have substantially different levels of identifiability. I use Tikhonov-type regularization together with a Morozov-type discrepancy principle to relate the regularization level to the observational error. Multi-start and multi-noise-level numerical experiments are used to study the robustness of the recovered parameters and to distinguish accurate state reconstruction from reliable identification of individual model coefficients.

This research forms the basis of the upcoming manuscript

> *Discretize-then-Optimize Parameter Estimation for Brain Tumor Growth Models using Tensorial ROMs*
>
> A. Islam, P. Amiri, M. Olshanskii, and A. Mang

Tensorial Reduced-Order Models for Inverse Problems

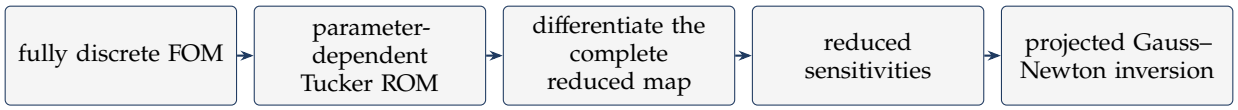
The next stage of my work brings the forward-surrogate and inverse-problem components together. Although a tensorial ROM can accelerate one forward evaluation dramatically, using it inside an optimization algorithm requires substantially more than replacing the high-fidelity solver by a faster approximation.

An important feature of the Tucker formulation I study is that the online reduced space is parameter dependent. For a new parameter vector θ, contractions of the parameter modes of the Tucker tensor generate a local reduced representation. Thus, the reduced basis itself depends on the parameter:

$$V = V(\theta).$$

This dependence has important consequences for inverse problems. Differentiating the reduced parameter-to-state map requires accounting not only for the reduced state variables but also for the parameter dependence of the reduced basis. Ignoring these terms would lead to derivatives corresponding to a different reduced model from the one actually used in the forward computation.

I am therefore developing a derivative-consistent reduced inverse framework based on



The intrusive reduced model retains the numerical structure of the high-fidelity solver. The complete parameter-dependent reduced map is differentiated, including contributions from the reduced basis, projected initial data, nonlinear reaction terms, and parameter-dependent diffusion operators.

The resulting reduced sensitivities can then be used to construct a Gauss–Newton system in the original physical parameter space. This permits the reduced and full-order inverse solvers to use

the same parameter bounds, scaling, regularization strategy, discrepancy principle, line-search mechanism, and stopping criteria.

The goal is not simply to obtain a fast optimizer, but to construct a reduced inverse method whose derivative information is mathematically consistent with the surrogate used to evaluate the objective. This distinction is important whenever reduced-order models are used inside optimization or statistical inference.

Correcting Reduced-Order Model Error in Tumor Growth Inversion

Reduced-order inversion introduces a second important question. Even when an optimization algorithm converges accurately for the reduced problem, the parameter recovered by the ROM need not coincide with the parameter that would be recovered by the full-order model. The difference is caused by surrogate-model discrepancy rather than optimization error.

My current collaborative research with Pegah Amiri and Andreas Mang investigates this problem using hyper-differential sensitivity analysis and post-optimality correction. Instead of repeating an expensive full-order inversion, we treat the difference between the full-order and reduced forward maps as a perturbation of the reduced optimization problem and quantify how the optimum changes under that perturbation.

For a reduced solution θ_{ROM}, the model discrepancy can be characterized schematically by

$$\delta(\theta) = u_{\mathrm{FOM}}(\theta) - u_{\mathrm{ROM}}(\theta).$$

Post-optimality sensitivity information then determines how this discrepancy changes the minimizer of the inverse problem. The resulting correction can be computed using reduced sensitivities and reduced Hessian information together with only a small number of expensive full-order evaluations.

I am studying both a direct linear correction and a continuation-based strategy in which the discrepancy is introduced gradually. The latter provides a mechanism for handling regimes in which a single linearized correction is insufficient.

The single-species tumor problem provides a controlled setting in which the corrected ROM parameter can be compared directly against a full-order inverse solution. The next stage is to transfer the methodology to the nine-parameter multi-species problem, where avoiding repeated full-order inversion provides a much larger computational benefit.

This work is being developed in the upcoming manuscript

> *Correcting Reduced-Order Model Error in Tumor Growth Inversion via Hyper-Differential Sensitivity Analysis*
>
> P. Amiri, A. Islam, and A. Mang

Together, my two ongoing inverse-problem projects address complementary questions: the first asks how to construct an efficient and derivative-consistent reduced inverse solver, while the second asks how to quantify and correct the parameter error that remains because the reduced model is only an approximation of the full-order dynamics.

Future Research Directions

My future research will continue to develop efficient and mathematically reliable algorithms for nonlinear parametric PDE systems. I am particularly interested in extending the ideas developed

during my doctoral research beyond tumor modeling and in connecting model reduction more closely with optimization, UQ, and scientific machine learning.

Reduced-Order PDE-Constrained Optimization. A primary direction is the development of reduced optimization methods in which parameter-dependent reduced spaces and their derivatives are incorporated consistently into gradient-, Gauss–Newton-, and Newton-type algorithms. Many nonlinear time-dependent PDE systems share the same basic computational difficulty as the tumor-growth problem: an individual simulation is manageable, but thousands of simulations required by optimization or inference are not. I am interested in developing reduced formulations that preserve derivative consistency while providing substantial online acceleration.

Model-Discrepancy-Aware Inverse Problems. My work on hyper-differential correction suggests a broader research direction in which surrogate error is treated explicitly as part of the inverse problem rather than ignored after model reduction. I plan to study error-aware reduced optimization algorithms, adaptive correction strategies, and methods for deciding when additional high-fidelity evaluations are necessary. An important objective is to establish practical relationships among surrogate accuracy, parameter-estimation accuracy, and computational cost.

Uncertainty Quantification and Statistical Inference. Deterministic parameter estimates alone do not characterize uncertainty caused by observational noise, incomplete data, parameter non-identifiability, and model discrepancy. I plan to combine low-rank tensor surrogates with Bayesian and statistical inference methods so that posterior exploration can be performed at a computational cost substantially below that of direct full-order sampling.

A particularly important question is how approximation errors introduced by a surrogate propagate into posterior parameter uncertainty. My work on deterministic model-error correction provides a natural starting point for developing uncertainty-aware reduced inverse methods.

Adaptive Tensor Approximation and Tensor Completion. The principal limitation of tensor-product surrogate construction is the rapid growth of the required training set as the parameter dimension increases. I plan to investigate tensor completion, cross approximation, randomized low-rank algorithms, and adaptive sampling strategies that avoid evaluating the full tensor-product grid.

Rather than computing every entry of a high-dimensional snapshot tensor, these methods can concentrate expensive high-fidelity simulations in regions where the solution changes rapidly or where parameter sensitivities indicate that additional resolution is important. Connecting adaptive snapshot acquisition directly to inverse-problem information is a particularly interesting direction.

High-Performance Scientific Computing. Large-scale reduced-order modeling requires both efficient algorithms and scalable implementations. The offline stage naturally involves large ensembles of independent or weakly coupled high-fidelity simulations and therefore provides significant opportunities for parallel computing.

I plan to continue developing implementations that combine parallel snapshot generation, sparse numerical linear algebra, randomized decompositions, memory-efficient tensor operations, and scalable iterative solvers. Such computational tools will enable the study of finer spatial meshes, higher-dimensional parameter spaces, three-dimensional multi-species models, and more complex coupled PDE systems.

Scientific Machine Learning and Hybrid Reduced Models. I am also interested in combining physics-based reduced models with data-driven approximation. Machine-learning methods may be

used to approximate reduced coordinates, identify regions of parameter space requiring additional training, accelerate nonlinear reduced operators, or learn corrections to imperfect surrogate models.

My objective is not to replace governing equations by black-box models, but to develop hybrid methods in which data-driven flexibility complements the interpretability and mathematical structure of numerical PDE models.

Patient-Specific Modeling and Digital Twins. My longer-term research goal is to integrate forward simulation, reduced-order modeling, inverse parameter estimation, model-error correction, and uncertainty quantification into computational pipelines for patient-specific biomedical modeling.

In such a framework, observational or imaging data would inform patient-specific parameters, reduced models would permit repeated simulation and inference, and uncertainty estimates would quantify confidence in the resulting predictions. Achieving this goal requires progress in parameter identifiability, incomplete-data inversion, model discrepancy, surrogate reliability, and scalable inference—all of which are natural extensions of my current research.

Research Vision

My broader objective is to develop computational methods that are simultaneously efficient, mathematically consistent, and reliable for optimization and inference. I am particularly interested in problems where numerical PDEs, low-rank approximation, optimization, and observational data interact and where computational efficiency cannot be separated from questions of approximation error and derivative accuracy.

My doctoral work has progressed from the construction of fast tensorial forward surrogates, to deterministic PDE-constrained parameter estimation, to derivative-consistent reduced inversion, and now to the correction of reduced-model error through hyper-differential sensitivity analysis. These projects form a unified research program centered on making complex nonlinear PDE models computationally useful in many-query settings.

By combining numerical analysis, tensor-based reduced-order modeling, PDE-constrained optimization, inverse problems, uncertainty quantification, scientific machine learning, and high-performance computing, I aim to develop scalable computational methods that can be applied not only to biomedical modeling but also to broader nonlinear parameter-dependent systems across computational science and engineering.